

Model Question Paper: Mathematics for AI

Course: Mathematics for Artificial Intelligence

Duration: 1 Hour (Quiz Section)

Max Marks: 20

Section 1: Quiz (Detailed Solutions)

Instructions: All questions are compulsory. Each question carries 2 marks.

1.1 Fundamental Subspaces

Question: If the column space of an 8×4 matrix A is 3-dimensional, give the dimensions of the other three fundamental subspaces.

Answer:

- **Column Space** $C(A)$: Dim = 3 (Given)
- **Row Space** $C(A^T)$: Dim = 3
- **Null Space** $N(A)$: Dim = 1
- **Left Null Space** $N(A^T)$: Dim = 5

Justification: Given a matrix A of size $m \times n$ where $m = 8$ (rows) and $n = 4$ (columns):

1. **Rank (r):** The dimension of the column space is the rank of the matrix. Thus, $r = 3$.
2. **Row Space:** The dimension of the row space is always equal to the rank. Dim $C(A^T) = r = 3$.
3. **Null Space:** The dimension of the null space is $n - r$ (Rank-Nullity Theorem). Dim $N(A) = 4 - 3 = 1$.
4. **Left Null Space:** The dimension of the left null space is $m - r$. Dim $N(A^T) = 8 - 3 = 5$.

1.2 Vector Spaces

Question: Assert or Reject the following statement with justification: *Statement: The set of all even degree polynomials (including 0) is a vector space.*

Answer: Reject.

Justification: A vector space must be closed under vector addition. Let V be the set of polynomials with even degree. Consider two polynomials in this set:

1. $P(x) = x^2 + x$ (Degree is 2, which is even)
2. $Q(x) = -x^2$ (Degree is 2, which is even)

If we add them:

$$P(x) + Q(x) = (x^2 + x) + (-x^2) = x$$

The result, x , has a degree of 1 (which is odd). Since the sum of two vectors in the set resulted in a vector outside the set, the set is **not closed under addition** and therefore is not a vector space.

(Note: If the question referred to "Even Functions" i.e., $f(-x) = f(x)$, then the answer would be Assert. However, "Even Degree" refers to the highest power, making the statement false).

1.3 Euclidean Norms & Dot Product

Question: Given the Euclidean norms $\|u\| = 4$, $\|v\| = 3$, and $\|u + v\| = 7$, then what is the dot product $u \cdot v$?

Answer: 12

Justification: We use the expansion of the squared norm:

$$\|u + v\|^2 = (u + v) \cdot (u + v) = \|u\|^2 + \|v\|^2 + 2(u \cdot v)$$

Substitute the given values:

$$7^2 = 4^2 + 3^2 + 2(u \cdot v)$$

$$49 = 16 + 9 + 2(u \cdot v)$$

$$49 = 25 + 2(u \cdot v)$$

$$24 = 2(u \cdot v)$$

$$u \cdot v = 12$$

Alternative Logic: Since $\|u + v\| = \|u\| + \|v\|$ ($7 = 4 + 3$), the Triangle Inequality becomes an equality. This implies vectors u and v are in the exact same direction. Therefore, the angle $\theta = 0^\circ$.

$$u \cdot v = \|u\|\|v\|\cos(0) = 4 \times 3 \times 1 = 12$$

1.4 Eigenvalues of Symmetric Matrices

Question: Assert or Reject the following statement with justification: *Statement: There exists a real symmetric matrix whose characteristic equation is $\lambda(\lambda^2 + 4)(\lambda - 1) = 0$.*

Answer: Reject.

Justification: The characteristic equation is $\lambda(\lambda^2 + 4)(\lambda - 1) = 0$. Solving for the roots (eigenvalues):

1. $\lambda = 0$
2. $\lambda - 1 = 0 \Rightarrow \lambda = 1$
3. $\lambda^2 + 4 = 0 \Rightarrow \lambda^2 = -4 \Rightarrow \lambda = \pm 2i$

The eigenvalues include imaginary numbers ($\pm 2i$). **Spectral Theorem Property:** All eigenvalues of a real symmetric matrix must be **real numbers**. Since this equation yields complex eigenvalues, no such real symmetric matrix exists.

1.5 Statistics Definitions

Question: Define heteroscedasticity.

Answer: Heteroscedasticity refers to the circumstance in regression analysis where the variability (variance) of the error term (residuals) is **not constant** across all levels of the independent variable(s).

Visually, in a scatter plot of residuals vs. predicted values, heteroscedasticity appears as a cone or fan shape, where the data points spread out more widely as the value increases or decreases.

1.6 Least Squares

Question: How is the ordinary least squares (OLS) problem different from a weighted least squares (WLS) case?

Answer:

1. Assumption of Variance:

- **OLS** assumes **homoscedasticity**, meaning every observation provides equally reliable information (constant variance of errors, σ^2).
- **WLS** acknowledges **heteroscedasticity**, assuming some observations are more "noisy" (higher variance) than others.

2. Minimization Objective:

- **OLS** minimizes the sum of squared residuals: $\sum (y_i - \hat{y}_i)^2$.
- **WLS** minimizes the *weighted* sum: $\sum w_i (y_i - \hat{y}_i)^2$, where weights w_i are inversely proportional to the variance ($w_i = 1/\sigma_i^2$). This gives less influence to noisy data points.

1.7 Vector Calculus - Kinematics

Question: The position vector of point P is $r = (3 \cos t, 5 \sin t, 4 \cos t)$. What is the speed?

Answer: 5

Justification: Velocity $v(t)$ is the derivative of position $r(t)$:

$$r(t) = \langle 3 \cos t, 5 \sin t, 4 \cos t \rangle$$

$$v(t) = r'(t) = \langle -3 \sin t, 5 \cos t, -4 \sin t \rangle$$

Speed is the magnitude of the velocity vector $\|v(t)\|$:

$$\begin{aligned} \text{Speed} &= \sqrt{(-3 \sin t)^2 + (5 \cos t)^2 + (-4 \sin t)^2} \\ &= \sqrt{9 \sin^2 t + 25 \cos^2 t + 16 \sin^2 t} \end{aligned}$$

Combine $\sin^2 t$ terms:

$$\begin{aligned} &= \sqrt{(9 + 16) \sin^2 t + 25 \cos^2 t} \\ &= \sqrt{25 \sin^2 t + 25 \cos^2 t} \\ &= \sqrt{25(\sin^2 t + \cos^2 t)} \end{aligned}$$

Since $\sin^2 t + \cos^2 t = 1$:

$$\text{Speed} = \sqrt{25} = 5$$

1.8 Gradient

Question: Obtain the gradient of the function $f(x, y) = (x + y)^2$.

Answer: $\begin{bmatrix} 2(x + y) \\ 2(x + y) \end{bmatrix}$

Justification: The gradient ∇f is a vector of partial derivatives:

$$\nabla f = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right\rangle$$

Let $u = x + y$, so $f = u^2$.

1. **Partial w.r.t x :** Using Chain Rule: $\frac{\partial f}{\partial x} = \frac{df}{du} \cdot \frac{\partial u}{\partial x} = 2u \cdot 1 = 2(x + y)$.
2. **Partial w.r.t y :** Using Chain Rule: $\frac{\partial f}{\partial y} = \frac{df}{du} \cdot \frac{\partial u}{\partial y} = 2u \cdot 1 = 2(x + y)$.

Thus, $\nabla f = [2(x + y), 2(x + y)]^T$.

1.9 Pseudoinverse

Question: Consider a 2×3 real matrix A , with rank of $A = 2$. What is the pseudoinverse of A ?

Answer: $A^+ = A^T(AA^T)^{-1}$

Justification: The matrix A has dimensions $m \times n = 2 \times 3$. Since the rank is 2 (which equals m , the number of rows), A has **Full Row Rank**.

For a matrix with full row rank, the Right Inverse exists. The formula for the Moore-Penrose pseudoinverse in this specific case is:

$$A^+ = A^T(AA^T)^{-1}$$

(Note: If it had full column rank, the formula would be $(A^T A)^{-1} A^T$. Since rows < columns here, it is the former).

1.10 Gradient Descent

Question: What could be the consequence of higher learning rate in gradient descent algorithm?

Answer:

1. **Overshooting:** The algorithm may take steps that are too large, jumping over the global minimum.
2. **Divergence:** Instead of minimizing the loss function, the loss may actually increase with each step, causing the algorithm to diverge towards infinity.
3. **Oscillation:** The parameters may bounce back and forth around the minimum without ever

Model Question Paper: Mathematics for AI (Part B)

Section: Part B (Long Answer & Conceptual) **Max Marks:** 16

Question 2(a)

Problem: Let $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$. Find out all the 4 fundamental subspaces associated with A .

Mention one basis each for each of the subspaces. (10 Marks)

Solution

Given matrix A of size 3×2 ($m = 3, n = 2$):

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$$

Step 1: Gaussian Elimination to find Rank (r) Performing row operations:

$$R_3 \rightarrow R_3 - (R_1 + R_2)$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} \xrightarrow{R_3 - R_1} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{bmatrix} \xrightarrow{R_3 - R_2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} = U$$

Since there are 2 pivots, the **Rank** $r = 2$.

1. Column Space $C(A)$

The column space contains all linear combinations of the columns of A . The basis is formed by the pivot columns of the original matrix A .

- **Dimension:** $r = 2$
- **Basis:** The first and second columns of A .

$$\text{Basis for } C(A) = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}$$

2. Row Space $C(A^T)$

The row space contains all linear combinations of the rows of A . The basis is formed by the non-zero rows of the reduced row echelon form (U).

- **Dimension:** $r = 2$
- **Basis:** The non-zero rows of U (transposed to vector format).

$$\text{Basis for } C(A^T) = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$$

3. Null Space $N(A)$

The null space contains solutions to $Ax = 0$.

- **Dimension:** $n - r = 2 - 2 = 0$.
- **Calculation:**

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \implies \begin{cases} x_1 = 0 \\ x_2 = 0 \end{cases}$$

The only solution is the zero vector.

- **Basis:** The empty set $\emptyset$ (or simply stated as having no basis vectors, just the zero vector).

4. Left Null Space $N(A^T)$

The left null space contains solutions to $A^T y = 0$.

- **Dimension:** $m - r = 3 - 2 = 1$.
- **Calculation:** Transpose A :

$$A^T = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

Solve $A^T y = 0$:

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Equations:

1. $y_1 + y_3 = 0 \implies y_1 = -y_3$
2. $y_2 + y_3 = 0 \implies y_2 = -y_3$

Let the free variable $y_3 = 1$. Then $y_1 = -1$ and $y_2 = -1$.

- **Basis:**

$$\text{Basis for } N(A^T) = \left\{ \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} \right\}$$

Question 2(b)

Problem: Assert or Reject the following statements with justification. (6 Marks)

Statement 1

"For a given $n \times n$ real matrix A , two eigenvalues can have the same eigenvectors."

Answer: Reject

Justification: By definition, an eigenvector v corresponding to an eigenvalue λ must satisfy $Av = \lambda v$ and $v \neq 0$. Suppose a single vector v is an eigenvector for two distinct eigenvalues λ_1 and λ_2 .

$$Av = \lambda_1 v \quad \text{and} \quad Av = \lambda_2 v$$

Subtracting the two equations gives:

$$\lambda_1 v - \lambda_2 v = 0 \implies (\lambda_1 - \lambda_2)v = 0$$

Since v is an eigenvector, $v \neq 0$. Therefore, it must be that $(\lambda_1 - \lambda_2) = 0$, which implies $\lambda_1 = \lambda_2$. Thus, distinct eigenvalues cannot share the same eigenvector. A specific direction (eigenvector) can only stretch by a single factor (eigenvalue).

(Note: While multiple eigenvectors can share the same eigenvalue—degenerate cases—the reverse is mathematically impossible).

Statement 2

" $\mathbb{R}^2$ is a subspace of $\mathbb{R}^3$."

Answer: Reject

Justification: Strictly speaking, a subspace must be a **subset** of the original vector space.

- Elements of $\mathbb{R}^2$ are 2-tuples: (x, y) .
- Elements of $\mathbb{R}^3$ are 3-tuples: (x, y, z) .

Because 2-tuples are not the same mathematical objects as 3-tuples, $\mathbb{R}^2$ is not contained inside $\mathbb{R}^3$. While $\mathbb{R}^3$ has subspaces that are **isomorphic** to $\mathbb{R}^2$ (for example, the plane $z = 0$, which is the set $\{(x, y, 0) \mid x, y \in \mathbb{R}\}$), $\mathbb{R}^2$ itself is not a subspace of $\mathbb{R}^3$.

Model Question Paper: Mathematics for AI (Part B - Q3)

Section: Part B (continued) **Max Marks:** 16

Question 3(a)

Problem: Given the following vectors:

$$u_1 = (3, -3, 0), \quad u_2 = (2, 2, -1), \quad u_3 = (1, 1, 4), \quad x = (5, -3, 1)$$

(i) Are the vectors u_1, u_2, u_3 orthogonal? (ii) Are the vectors u_1, u_2, u_3 orthonormal? (iii) Express the vector x as a linear combination of the vectors u_1, u_2, u_3 . (Marks: 2 + 1 + 5 = 8)

Solution

(i) Checking for Orthogonality Vectors are orthogonal if their dot product is zero. We check all pairs:

$$1. \quad u_1 \cdot u_2 = (3)(2) + (-3)(2) + (0)(-1) = 6 - 6 + 0 = 0$$

$$2. \quad u_1 \cdot u_3 = (3)(1) + (-3)(1) + (0)(4) = 3 - 3 + 0 = 0$$

$$3. \quad u_2 \cdot u_3 = (2)(1) + (2)(1) + (-1)(4) = 2 + 2 - 4 = 0$$

Conclusion: Since all pairwise dot products are zero, the vectors u_1, u_2, u_3 are orthogonal.

(ii) Checking for Orthonormality Vectors are orthonormal if they are orthogonal AND their magnitudes (norms) are all equal to 1. Let's check the norms:

$$\bullet \quad \|u_1\|^2 = 3^2 + (-3)^2 + 0^2 = 9 + 9 = 18 \neq 1$$

$$\bullet \quad \|u_2\|^2 = 2^2 + 2^2 + (-1)^2 = 4 + 4 + 1 = 9 \neq 1$$

$$\bullet \quad \|u_3\|^2 = 1^2 + 1^2 + 4^2 = 1 + 1 + 16 = 18 \neq 1$$

Conclusion: The vectors are not orthonormal because they are not unit vectors.

(iii) Expressing x as a Linear Combination Since $\{u_1, u_2, u_3\}$ is an orthogonal basis, we can find the coefficients c_1, c_2, c_3 for $x = c_1u_1 + c_2u_2 + c_3u_3$ using the formula:

$$c_i = \frac{x \cdot u_i}{u_i \cdot u_i} = \frac{x \cdot u_i}{\|u_i\|^2}$$

Calculate Coefficients:

1. For c_1 :

$$\bullet \quad x \cdot u_1 = (5)(3) + (-3)(-3) + (1)(0) = 15 + 9 + 0 = 24$$

$$\bullet \quad \|u_1\|^2 = 18$$

$$\bullet \quad c_1 = \frac{24}{18} = \frac{4}{3}$$

2. For c_2 :

- $x \cdot u_2 = (5)(2) + (-3)(2) + (1)(-1) = 10 - 6 - 1 = 3$
- $\|u_2\|^2 = 9$
- $c_2 = \frac{3}{9} = \frac{1}{3}$

3. For c_3 :

- $x \cdot u_3 = (5)(1) + (-3)(1) + (1)(4) = 5 - 3 + 4 = 6$
- $\|u_3\|^2 = 18$
- $c_3 = \frac{6}{18} = \frac{1}{3}$

Final Expression:

$$x = \frac{4}{3}u_1 + \frac{1}{3}u_2 + \frac{1}{3}u_3$$

Question 3(b)

(i) If P is a projection matrix, prove that $P^2 = P$. (Marks: 4)

Proof

Conceptual Approach: Geometrically, a projection matrix P projects any vector b onto a subspace. Let Pb be the projection of vector b . Since Pb is already in the subspace, projecting it again should not change it. Therefore, $P(Pb) = Pb$, which implies $P^2 = P$.

Algebraic Proof: The formula for a projection matrix onto the column space of a matrix A (where A has independent columns) is:

$$P = A(A^T A)^{-1} A^T$$

Let's calculate P^2 :

$$P^2 = P \cdot P = (A(A^T A)^{-1} A^T) (A(A^T A)^{-1} A^T)$$

Associating the middle terms:

$$P^2 = A(A^T A)^{-1} \underbrace{(A^T A)(A^T A)^{-1}}_I A^T$$

Since $(A^T A)(A^T A)^{-1} = I$ (Identity Matrix):

$$P^2 = A(A^T A)^{-1} I A^T$$

$$P^2 = A(A^T A)^{-1} A^T$$

By observation, the right side is exactly the definition of P .

$$P^2 = P$$

Hence Proved.

(ii) Obtain the matrix that projects every vector in $\mathbb{R}^2$ along the line passing through the origin and in the direction of the vector $u = (1, 1)$. (Marks: 4)

Solution

The projection matrix P onto a line spanned by a vector a is given by the formula:

$$P = \frac{aa^T}{a^T a}$$

Given vector $a = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

1. Calculate Numerator (aa^T):

$$aa^T = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \end{bmatrix} = \begin{bmatrix} 1(1) & 1(1) \\ 1(1) & 1(1) \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

2. Calculate Denominator ($a^T a$):

$$a^T a = \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 1(1) + 1(1) = 2$$

3. Construct P :

$$P = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix}$$

Answer: The projection matrix is $\begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix}$.

Model Question Paper: Mathematics for AI (Part B - Q4)

Section: Part B (continued) **Max Marks:** 16

Question 4(a)

Problem: Obtain the orthogonal complement subspace of the plane $x_1 + x_2 + x_3 = 0$. (6 marks)

Solution

1. Identify the Subspace: The given equation $x_1 + x_2 + x_3 = 0$ represents a plane passing through the origin in $\mathbb{R}^3$. Let this subspace be V . Any vector $x = (x_1, x_2, x_3)$ in this plane satisfies the condition that its dot product with the normal vector is zero.

The equation can be written as a dot product:

$$\begin{bmatrix} 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

2. Relate to Fundamental Subspaces: If we consider the matrix $A = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}$, then the subspace V is exactly the **Null Space** of A ($N(A)$).

By the Fundamental Theorem of Linear Algebra, the orthogonal complement of the Null Space of A is the **Row Space** of A ($C(A^T)$).

$$V^\perp = (N(A))^\perp = C(A^T)$$

3. Find the Basis of the Complement: The row space of A is spanned by the row vector(s) of A .

$$\text{Row Vector} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

Thus, the orthogonal complement $V^\perp$ is the line (1D subspace) passing through the origin in the direction of the vector $(1, 1, 1)$.

Answer: The orthogonal complement subspace is the line spanned by the vector:

$$\text{Basis}(V^\perp) = \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}$$

Question 4(b)

Problem: Consider the following data matrix: $X = \begin{bmatrix} 4 & 1 \\ 2 & 3 \\ 5 & 4 \\ 1 & 0 \end{bmatrix}$. We want to represent the data in

only one dimension. Compute the unit-length principal component directions of X , and state which one the PCA algorithm would choose if you request just one principal component. Obtain the projections of all four sample points onto the principal direction. (10 marks)

Solution

Step 1: Center the Data (Mean Subtraction) First, calculate the mean of each feature (column).

- $\bar{x}_1 = \frac{4+2+5+1}{4} = \frac{12}{4} = 3$
- $\bar{x}_2 = \frac{1+3+4+0}{4} = \frac{8}{4} = 2$

Subtract the means from the original data matrix X to get the centered matrix B :

$$B = \begin{bmatrix} 4-3 & 1-2 \\ 2-3 & 3-2 \\ 5-3 & 4-2 \\ 1-3 & 0-2 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ -1 & 1 \\ 2 & 2 \\ -2 & -2 \end{bmatrix}$$

Step 2: Compute the Covariance Matrix We can calculate the covariance matrix Σ (or simply analyze $B^T B$).

$$S = B^T B = \begin{bmatrix} 1 & -1 & 2 & -2 \\ -1 & 1 & 2 & -2 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -1 & 1 \\ 2 & 2 \\ -2 & -2 \end{bmatrix}$$

$$S = \begin{bmatrix} 1+1+4+4 & -1-1+4+4 \\ -1-1+4+4 & 1+1+4+4 \end{bmatrix} = \begin{bmatrix} 10 & 6 \\ 6 & 10 \end{bmatrix}$$

Step 3: Find Eigenvalues and Eigenvectors Find the eigenvalues λ of S by solving $\det(S - \lambda I) = 0$:

$$\det \begin{bmatrix} 10-\lambda & 6 \\ 6 & 10-\lambda \end{bmatrix} = 0$$

$$(10-\lambda)^2 - 36 = 0$$

$$10-\lambda = \pm 6 \implies \lambda = 16, \lambda = 4$$

The eigenvalues are $\lambda_1 = 16$ and $\lambda_2 = 4$.

Step 4: Find Unit Eigenvectors (Principal Directions)

- For $\lambda_1 = 16$:

$$\begin{bmatrix} 10 - 16 & 6 \\ 6 & 10 - 16 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \implies \begin{bmatrix} -6 & 6 \\ 6 & -6 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = 0$$

Equation: $-6u_1 + 6u_2 = 0 \implies u_1 = u_2$. Normalized vector:

$$v_1 = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} \approx \begin{bmatrix} 0.707 \\ 0.707 \end{bmatrix}.$$

- For $\lambda_2 = 4$:

$$\begin{bmatrix} 10 - 4 & 6 \\ 6 & 10 - 4 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \implies \begin{bmatrix} 6 & 6 \\ 6 & 6 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 0$$

Equation: $6v_1 + 6v_2 = 0 \implies v_1 = -v_2$. Normalized vector:

$$v_2 = \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix} \approx \begin{bmatrix} 0.707 \\ -0.707 \end{bmatrix}.$$

Choice of Principal Component: PCA chooses the direction with the highest variance (largest eigenvalue). Since $\lambda_1 = 16 > \lambda_2 = 4$, the chosen direction is v_1 .

Chosen Direction: $\frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

Step 5: Projections The projection of the data points onto the 1D principal component space is given by the dot product of the centered data B with the chosen unit eigenvector v_1 .

$$z = Bv_1$$

1. **Point 1** (1, -1): $1(0.707) + (-1)(0.707) = 0$
2. **Point 2** (-1, 1): $-1(0.707) + 1(0.707) = 0$
3. **Point 3** (2, 2): $2(0.707) + 2(0.707) = 2.828$ (Exact: $2\sqrt{2}$)
4. **Point 4** (-2, -2): $-2(0.707) + (-2)(0.707) = -2.828$ (Exact: $-2\sqrt{2}$)

Final Result (1D Representation): The projections (scores) are:

$$Z = \begin{bmatrix} 0 \\ 0 \\ 2\sqrt{2} \\ -2\sqrt{2} \end{bmatrix}$$

Model Question Paper: Mathematics for AI (Part B - Q5)

Section: Part B (continued) **Max Marks:** 16

Question 5(a)

Problem: Suppose you measure your pulse rate x three times and get values $x = b_1, x = b_2, x = b_3$. If each right side b_i are independent and has zero mean and variances $\sigma_1^2 = 1/16, \sigma_2^2 = 1/9, \sigma_3^2 = 1/36$, find the best estimate of $\hat{x}$ based on b_1, b_2, b_3 . (10 marks)

Solution

This is a **Weighted Least Squares (WLS)** problem. Since the measurements have different variances (reliabilities), we cannot simply take the arithmetic mean. We must give more weight to the measurements with lower variance (higher precision).

Step 1: Determine the Weights The weight w_i for each observation is inversely proportional to its variance σ_i^2 .

$$w_i = \frac{1}{\sigma_i^2}$$

- **Weight for b_1 :**

$$w_1 = \frac{1}{1/16} = 16$$

- **Weight for b_2 :**

$$w_2 = \frac{1}{1/9} = 9$$

- **Weight for b_3 :**

$$w_3 = \frac{1}{1/36} = 36$$

Step 2: Apply the WLS Estimation Formula For a scalar parameter x estimated from multiple observations b_i , the weighted least squares estimate $\hat{x}$ minimizes the weighted sum of squared errors:

$$J = \sum_{i=1}^3 w_i (x - b_i)^2$$

The solution to this minimization is the **weighted average**:

$$\hat{x} = \frac{\sum_{i=1}^3 w_i b_i}{\sum_{i=1}^3 w_i}$$

Step 3: Calculate the Estimate Numerator:

$$\sum w_i b_i = 16b_1 + 9b_2 + 36b_3$$

Denominator:

$$\sum w_i = 16 + 9 + 36 = 61$$

Final Answer: The best estimate is:

$$\hat{x} = \frac{16b_1 + 9b_2 + 36b_3}{61}$$

Question 5(b)

Problem: List out 3 limitations of Gaussian Mixture Models (GMM). (6 marks)

Solution

1. Sensitivity to Initialization (Local Minima) GMMs are typically trained using the Expectation-Maximization (EM) algorithm. EM is a local optimization method, meaning it is highly sensitive to the initial values of the parameters (means, covariances, and mixing coefficients). If initialized poorly, the algorithm may converge to a suboptimal local minimum rather than the global maximum likelihood, leading to poor clustering results.

2. Assumption of Convex/Elliptical Shapes GMMs assume that the data points in each cluster follow a Gaussian distribution. This means the model inherently assumes that clusters are elliptical (or spherical) convex shapes. GMMs often struggle to model clusters with complex, non-convex geometric shapes (e.g., crescent shapes, spirals, or arbitrary manifolds), where density-based methods like DBSCAN might perform better.

3. Need to Specify the Number of Components (k) Unlike some non-parametric methods, GMM requires the user to specify the number of mixture components (clusters) k in advance. Determining the optimal k is not automatic and often requires model selection criteria such as the Bayesian Information Criterion (BIC) or Akaike Information Criterion (AIC), which adds computational complexity and uncertainty to the modeling process.

Model Question Paper: Mathematics for AI (Part B - Q6)

Section: Part B (continued) **Max Marks:** 16

Question 6(a)

Problem: List out 4 limitations of Expectation Maximization (EM) algorithm. (12 marks)

Solution

1. Sensitivity to Initialization (Local Optima) The EM algorithm is a deterministic local optimization method. It guarantees that the likelihood increases with each iteration, but it converges to a **local maximum** of the likelihood function, not necessarily the global maximum. Consequently, the final solution depends heavily on the initial parameter values. If initialized in a "bad" region of the parameter space, the algorithm may get stuck in a suboptimal solution. To mitigate this, multiple random initializations are often used.

2. Slow Convergence Rate The convergence of the EM algorithm can be very slow, particularly when the amount of missing information is large or when the mixture components are not well-separated (high overlap). The rate of convergence is typically linear, meaning the error decreases by a roughly constant factor at each step. In the vicinity of the optimum, the steps can become vanishingly small, requiring many iterations to reach a strict tolerance level.

3. Assumption of Specific Distributions EM requires a specific probabilistic model (e.g., Gaussian Mixture Model) to be defined beforehand. It assumes that the data is generated from a specific family of distributions. If the actual data distribution does not match the assumed model (for example, fitting spherical Gaussians to data that is actually non-convex or manifold-like), the algorithm will yield poor results regardless of convergence.

4. Computational Complexity in High Dimensions For high-dimensional data or complex models, the E-step (Expectation) and M-step (Maximization) can become computationally expensive.

- **E-step:** May require integrating over high-dimensional latent variables, which is often intractable and requires approximation methods (like Variational EM).
- **M-step:** May require inverting large covariance matrices (scaling cubically with dimension, $O(d^3)$), which is costly and numerically unstable if the matrices are singular.

Question 6(b)

Problem: List 4 applications of Kalman filters. (4 marks)

Solution

1. Navigation and Guidance Systems (GPS/INS Integration) Kalman filters are the standard for fusing noisy data from different sensors, such as GPS (Global Positioning System) and INS (Inertial Navigation Systems). They are used in aircraft, autonomous vehicles, submarines, and

missiles to estimate precise position, velocity, and orientation by correcting the drift of inertial sensors with periodic GPS updates.

2. Computer Vision and Object Tracking In video processing, Kalman filters are used to track moving objects (e.g., pedestrians, vehicles, or balls in sports analytics) across frames. They predict the object's future location based on its current trajectory, allowing the system to maintain tracking even if the object is briefly occluded or if the detection is noisy.

3. Robotics (SLAM) Kalman filters (specifically the Extended Kalman Filter or EKF) are fundamental in robotics for Simultaneous Localization and Mapping (SLAM). They help a robot estimate its own pose (position and orientation) within a map while simultaneously building or updating that map based on noisy sensor readings from LiDAR or cameras.

4. Economic and Financial Modeling Economists use Kalman filters for time-series analysis to estimate unobservable variables (latent states) from observable market data. Applications include estimating the "true" inflation rate, modeling time-varying parameters in regression models, or smoothing noisy stock price data to identify underlying trends.

Model Question Paper: Mathematics for AI (Part B - Q7)

Section: Part B (continued) **Max Marks:** 16

Question 7

Problem Statement: The quadratic form $x^T Ax$ is a form we will encounter often. In this question, we are interested in $\frac{d(x^T Ax)}{dx}$. Assume that A is not a function of x .

(a) Expansion and Naming

Evaluate $x^T Ax$ when $x = [x_1, x_2]^T$ and the (i, j) -th element of A is A_{ij} . Why do you think $x^T Ax$ is called the quadratic form?

Solution: Let $A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$ and $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$.

First, compute Ax :

$$Ax = \begin{bmatrix} A_{11}x_1 + A_{12}x_2 \\ A_{21}x_1 + A_{22}x_2 \end{bmatrix}$$

Next, compute the dot product $x^T(Ax)$:

$$\begin{aligned} x^T Ax &= \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} A_{11}x_1 + A_{12}x_2 \\ A_{21}x_1 + A_{22}x_2 \end{bmatrix} \\ &= x_1(A_{11}x_1 + A_{12}x_2) + x_2(A_{21}x_1 + A_{22}x_2) \\ &= A_{11}x_1^2 + A_{12}x_1x_2 + A_{21}x_2x_1 + A_{22}x_2^2 \end{aligned}$$

Why it is called "Quadratic Form": It is called a quadratic form because the resulting scalar expression is a polynomial where **every term has a degree of exactly 2**. The terms are either squared variables (x_1^2, x_2^2) or cross-products (x_1x_2).

(b) Definition of Derivative

Which definition of the derivative do we need in order to evaluate $\frac{d(x^T Ax)}{dx}$?

Solution: Since $f(x) = x^T Ax$ is a scalar-valued function ($f : \mathbb{R}^n \rightarrow \mathbb{R}$) taking a vector input x , we need the definition of the **Gradient Vector** (or Vector Derivative).

The derivative is defined as the vector of partial derivatives with respect to each component of x :

$$\frac{df}{dx} = \nabla f = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{bmatrix}$$

(c) Evaluation in $\mathbb{R}^2$

Assume $x \in \mathbb{R}^2$ and $A \in \mathbb{R}^{2 \times 2}$. Evaluate $\frac{d(x^T Ax)}{dx}$.

Solution: Let $f(x) = A_{11}x_1^2 + (A_{12} + A_{21})x_1x_2 + A_{22}x_2^2$.

Compute the partial derivatives:

1. **With respect to x_1 :**

$$\begin{aligned} \frac{\partial f}{\partial x_1} &= \frac{\partial}{\partial x_1}(A_{11}x_1^2) + \frac{\partial}{\partial x_1}((A_{12} + A_{21})x_1x_2) + 0 \\ &= 2A_{11}x_1 + (A_{12} + A_{21})x_2 \end{aligned}$$

2. **With respect to x_2 :**

$$\begin{aligned} \frac{\partial f}{\partial x_2} &= 0 + \frac{\partial}{\partial x_2}((A_{12} + A_{21})x_1x_2) + \frac{\partial}{\partial x_2}(A_{22}x_2^2) \\ &= (A_{12} + A_{21})x_1 + 2A_{22}x_2 \end{aligned}$$

Resulting Vector:

$$\nabla f = \begin{bmatrix} 2A_{11}x_1 + A_{12}x_2 + A_{21}x_2 \\ A_{12}x_1 + A_{21}x_1 + 2A_{22}x_2 \end{bmatrix}$$

This can be rewritten as the sum of Ax and $A^T x$:

$$= \begin{bmatrix} A_{11}x_1 + A_{12}x_2 \\ A_{21}x_1 + A_{22}x_2 \end{bmatrix} + \begin{bmatrix} A_{11}x_1 + A_{21}x_2 \\ A_{12}x_1 + A_{22}x_2 \end{bmatrix} = Ax + A^T x$$

(d) Generalization to $\mathbb{R}^n$

Generalize the previous result to when $x \in \mathbb{R}^n$ and $A \in \mathbb{R}^{n \times n}$ and evaluate $\frac{d(x^T Ax)}{dx}$. Can you express the result in matrix form?

Solution: Using the product rule for matrix calculus or the summation notation $\sum_{i,j} A_{ij}x_i x_j$:

The derivative of the quadratic form $x^T Ax$ with respect to the vector x is:

$$\frac{d(x^T Ax)}{dx} = (A + A^T)x$$

Proof (Matrix Form):

$$d(x^T Ax) = d(x^T)Ax + x^T d(Ax)$$

$$= (dx)^T Ax + x^T A dx$$

Since the result is a scalar, $(dx)^T Ax = ((dx)^T Ax)^T = x^T A^T dx$.

$$= x^T A^T dx + x^T A dx$$

$$= x^T (A^T + A) dx$$

Thus, the gradient (column vector) is $(A + A^T)x$.

(e) Symmetric Matrix Case

What happens when A is a symmetric matrix, i.e., $A^T = A$?

Solution: If A is symmetric, then $A^T = A$. Substituting this into the general formula from (d):

$$\frac{d(x^T Ax)}{dx} = (A + A)x = 2Ax$$

Model Question Paper: Mathematics for AI (Part B - Q8)

Section: Part B (continued) **Max Marks:** 16

Question 8

Problem: Suppose the 2×1 function $\mathbf{f}$ of the 3×1 vector $\mathbf{x}$ is given by:

$$\mathbf{f}(\mathbf{x}) = \begin{pmatrix} x_1^2 + x_2^2 + x_3^2 \\ 2x_1 - x_2 - x_3 \end{pmatrix}$$

And the function $\mathbf{g}$ of the 2×1 vector $\mathbf{z}$ is given by:

$$\mathbf{g}(\mathbf{z}) = \begin{pmatrix} z_2 \\ z_1 \\ z_1 z_2 \end{pmatrix}$$

(Note: Based on the matrix structure in the problem image, $\mathbf{g}$ outputs a 3-component vector, making it a 3×1 function despite the text description).

Use the chain rule to compute:

$$\frac{\partial}{\partial \mathbf{x}^T} \mathbf{y}(\mathbf{x})$$

where $\mathbf{y}(\mathbf{x})$ is the composite function defined by $\mathbf{y}(\mathbf{x}) = \mathbf{g}(\mathbf{f}(\mathbf{x}))$.

Solution

1. Define the Variables and Functions

Let $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$. The inner function $\mathbf{z} = \mathbf{f}(\mathbf{x})$ maps $\mathbb{R}^3 \rightarrow \mathbb{R}^2$:

$$z_1 = x_1^2 + x_2^2 + x_3^2$$

$$z_2 = 2x_1 - x_2 - x_3$$

The outer function $\mathbf{y} = \mathbf{g}(\mathbf{z})$ maps $\mathbb{R}^2 \rightarrow \mathbb{R}^3$:

$$y_1 = z_2$$

$$y_2 = z_1$$

$$y_3 = z_1 z_2$$

2. State the Multivariate Chain Rule

The derivative $\frac{\partial}{\partial \mathbf{x}^T} \mathbf{y}(\mathbf{x})$ represents the Jacobian matrix of the composite function $\mathbf{y}$ with respect to $\mathbf{x}$. According to the chain rule:

$$J_y(\mathbf{x}) = J_g(\mathbf{z}) \cdot J_f(\mathbf{x})$$

Where:

- $J_f(\mathbf{x})$ is the 2×3 Jacobian matrix of the inner function $\mathbf{f}$.
- $J_g(\mathbf{z})$ is the 3×2 Jacobian matrix of the outer function $\mathbf{g}$.
- The result will be a 3×3 matrix.

3. Compute the Jacobian of the Inner Function (J_f)

We calculate the partial derivatives of z_1 and z_2 with respect to x_1, x_2, x_3 :

- **Row 1 (Derivatives of $z_1 = x_1^2 + x_2^2 + x_3^2$):**

$$\frac{\partial z_1}{\partial x_1} = 2x_1, \quad \frac{\partial z_1}{\partial x_2} = 2x_2, \quad \frac{\partial z_1}{\partial x_3} = 2x_3$$

- **Row 2 (Derivatives of $z_2 = 2x_1 - x_2 - x_3$):**

$$\frac{\partial z_2}{\partial x_1} = 2, \quad \frac{\partial z_2}{\partial x_2} = -1, \quad \frac{\partial z_2}{\partial x_3} = -1$$

$$J_f(\mathbf{x}) = \begin{pmatrix} 2x_1 & 2x_2 & 2x_3 \\ 2 & -1 & -1 \end{pmatrix}$$

4. Compute the Jacobian of the Outer Function (J_g)

We calculate the partial derivatives of y_1, y_2, y_3 with respect to z_1, z_2 :

- **Row 1 (Derivatives of $y_1 = z_2$):**

$$\frac{\partial y_1}{\partial z_1} = 0, \quad \frac{\partial y_1}{\partial z_2} = 1$$

- **Row 2 (Derivatives of $y_2 = z_1$):**

$$\frac{\partial y_2}{\partial z_1} = 1, \quad \frac{\partial y_2}{\partial z_2} = 0$$

- **Row 3 (Derivatives of $y_3 = z_1 z_2$):**

$$\frac{\partial y_3}{\partial z_1} = z_2, \quad \frac{\partial y_3}{\partial z_2} = z_1$$

$$J_g(\mathbf{z}) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \\ z_2 & z_1 \end{pmatrix}$$

5. Apply the Chain Rule (Matrix Multiplication)

$$\frac{\partial \mathbf{y}}{\partial \mathbf{x}^T} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \\ z_2 & z_1 \end{pmatrix} \begin{pmatrix} 2x_1 & 2x_2 & 2x_3 \\ 2 & -1 & -1 \end{pmatrix}$$

Let's compute the product row by row:

- **Row 1:** $[0, 1] \cdot \text{Cols of } J_f$

$$\begin{aligned} &= [0(2x_1) + 1(2), \quad 0(2x_2) + 1(-1), \quad 0(2x_3) + 1(-1)] \\ &= [2, -1, -1] \end{aligned}$$

- **Row 2:** $[1, 0] \cdot \text{Cols of } J_f$

$$\begin{aligned} &= [1(2x_1) + 0(2), \quad 1(2x_2) + 0(-1), \quad 1(2x_3) + 0(-1)] \\ &= [2x_1, 2x_2, 2x_3] \end{aligned}$$

- **Row 3:** $[z_2, z_1] \cdot \text{Cols of } J_f$

$$= [z_2(2x_1) + z_1(2), \quad z_2(2x_2) + z_1(-1), \quad z_2(2x_3) + z_1(-1)]$$

6. Final Substitution

Substitute z_1 and z_2 back into the expressions for Row 3 to get the result purely in terms of $\mathbf{x}$.

- $z_1 = x_1^2 + x_2^2 + x_3^2$
- $z_2 = 2x_1 - x_2 - x_3$

Row 3 Element 1:

$$\begin{aligned} &2x_1(2x_1 - x_2 - x_3) + 2(x_1^2 + x_2^2 + x_3^2) \\ &= 4x_1^2 - 2x_1x_2 - 2x_1x_3 + 2x_1^2 + 2x_2^2 + 2x_3^2 \end{aligned}$$

$$= 6x_1^2 + 2x_2^2 + 2x_3^2 - 2x_1x_2 - 2x_1x_3$$

Row 3 Element 2:

$$2x_2(2x_1 - x_2 - x_3) - 1(x_1^2 + x_2^2 + x_3^2)$$

$$= 4x_1x_2 - 2x_2^2 - 2x_2x_3 - x_1^2 - x_2^2 - x_3^2$$

$$= -x_1^2 - 3x_2^2 - x_3^2 + 4x_1x_2 - 2x_2x_3$$

Row 3 Element 3:

$$2x_3(2x_1 - x_2 - x_3) - 1(x_1^2 + x_2^2 + x_3^2)$$

$$= 4x_1x_3 - 2x_2x_3 - 2x_3^2 - x_1^2 - x_2^2 - x_3^2$$

$$= -x_1^2 - x_2^2 - 3x_3^2 + 4x_1x_3 - 2x_2x_3$$

Final Answer Matrix:

Model Question Paper: Mathematics for AI (Part B - Q9)

Section: Part B (continued) **Max Marks:** 16

Question 9(a)

Problem: The function $f(x, y, z) = x^2 + y^2 + 3z^2 - xy + 2xz + yz$ defined on $\mathbb{R}^3$ has only one stationary point. Show that it is a local minimum. (6 marks)

Solution

1. Find the Stationary Points A stationary point occurs where the gradient ∇f is zero. We compute the partial derivatives with respect to x , y , and z :

$$f_x = \frac{\partial f}{\partial x} = 2x - y + 2z$$

$$f_y = \frac{\partial f}{\partial y} = 2y - x + z$$

$$f_z = \frac{\partial f}{\partial z} = 6z + 2x + y$$

Set these to zero to form a system of linear equations:

1. $2x - y + 2z = 0$
2. $-x + 2y + z = 0$
3. $2x + y + 6z = 0$

Solving this system:

- From (1), multiply by 2: $4x - 2y + 4z = 0$. Add to (2):
 $(4x - 2y + 4z) + (-x + 2y + z) = 0 \implies 3x + 5z = 0 \implies x = -\frac{5}{3}z$.
- Substitute x into (3):
 $2(-\frac{5}{3}z) + y + 6z = 0 \implies -\frac{10}{3}z + y + \frac{18}{3}z = 0 \implies y + \frac{8}{3}z = 0 \implies y = -\frac{8}{3}z$.
- Substitute x and y back into (1): $2(-\frac{5}{3}z) - (-\frac{8}{3}z) + 2z = 0$
 $-\frac{10}{3}z + \frac{8}{3}z + \frac{6}{3}z = 0 \implies \frac{4}{3}z = 0 \implies z = 0$.

If $z = 0$, then $x = 0$ and $y = 0$. Thus, the **only stationary point is** $(0, 0, 0)$.

2. Analyze the Hessian Matrix To determine the nature of the stationary point, we compute the Hessian matrix H of second-order partial derivatives:

$$H = \begin{bmatrix} f_{xx} & f_{xy} & f_{xz} \\ f_{yx} & f_{yy} & f_{yz} \\ f_{zx} & f_{zy} & f_{zz} \end{bmatrix} = \begin{bmatrix} 2 & -1 & 2 \\ -1 & 2 & 1 \\ 2 & 1 & 6 \end{bmatrix}$$

3. Test for Positive Definiteness We check the leading principal minors (D_k) of H :

- $D_1: |2| = 2 > 0$
- $D_2: \begin{vmatrix} 2 & -1 \\ -1 & 2 \end{vmatrix} = (2)(2) - (-1)(-1) = 4 - 1 = 3 > 0$
- $D_3: \det(H) = 2(12 - 1) - (-1)(-6 - 2) + 2(-1 - 4) = 2(11) + 1(-8) + 2(-5) = 22 - 8 - 10 = 4 > 0$

Conclusion: Since all leading principal minors are positive ($D_1 > 0, D_2 > 0, D_3 > 0$), the Hessian matrix is **Positive Definite**. Therefore, the function is convex near the stationary point, and $(0, 0, 0)$ is a **Local Minimum**.

Question 9(b)

Problem: Find the least square solution to the following system of equations:

$$x_1 + x_2 = 2$$

$$x_1 - x_2 = 0$$

$$2x_1 + x_2 = -4$$

(10 marks)

Solution

This is an overdetermined system $Ax = b$.

1. Define Matrices

$$A = \begin{bmatrix} 1 & 1 \\ 1 & -1 \\ 2 & 1 \end{bmatrix}, \quad x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \quad b = \begin{bmatrix} 2 \\ 0 \\ -4 \end{bmatrix}$$

2. Form the Normal Equation The least squares solution $\hat{x}$ satisfies $A^T A \hat{x} = A^T b$.

Calculate $A^T A$:

$$A^T A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \\ 2 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1(1) + 1(1) + 2(2) & 1(1) + 1(-1) + 2(1) \\ 1(1) - 1(1) + 1(2) & 1(1) - 1(-1) + 1(1) \end{bmatrix}$$

$$= \begin{bmatrix} 6 & 2 \\ 2 & 3 \end{bmatrix}$$

Calculate $A^T b$:

$$A^T b = \begin{bmatrix} 1 & 1 & 2 \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \\ -4 \end{bmatrix}$$

$$= \begin{bmatrix} 1(2) + 1(0) + 2(-4) \\ 1(2) - 1(0) + 1(-4) \end{bmatrix}$$

$$= \begin{bmatrix} 2 - 8 \\ 2 - 4 \end{bmatrix} = \begin{bmatrix} -6 \\ -2 \end{bmatrix}$$

3. Solve for $\hat{x}$ We solve the system $\begin{bmatrix} 6 & 2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -6 \\ -2 \end{bmatrix}$.

From row 2: $2x_1 + 3x_2 = -2 \implies 2x_1 = -2 - 3x_2$. From row 1: $6x_1 + 2x_2 = -6$. Divide by 2: $3x_1 + x_2 = -3$.

Substitute $2x_1$ isn't clean, let's use Gaussian elimination or Cramer's rule. Multiply the second equation ($2x_1 + 3x_2 = -2$) by 3:

$$6x_1 + 9x_2 = -6$$

Subtract the first equation ($6x_1 + 2x_2 = -6$) from this:

$$(6x_1 + 9x_2) - (6x_1 + 2x_2) = -6 - (-6)$$

$$7x_2 = 0 \implies x_2 = 0$$

Substitute $x_2 = 0$ back into any equation:

$$2x_1 + 3(0) = -2 \implies 2x_1 = -2 \implies x_1 = -1$$

Model Question Paper: Mathematics for AI (Part B - Q10)

Section: Part B (continued) **Max Marks:** 16

Question 10(a)

Problem: Use the method of Lagrange multipliers to find the minimum value of $f(x, y) = x^2 + 4y^2 - 2x + 8y$ subject to the constraint $x + 2y = 7$. (16 marks)

Solution

1. Define the Functions

- **Objective Function:** We want to minimize $f(x, y) = x^2 + 4y^2 - 2x + 8y$.
- **Constraint Function:** The constraint is $x + 2y = 7$. We write this as $g(x, y) = x + 2y - 7 = 0$.

2. Formulate the Lagrange Equations According to the method of Lagrange multipliers, the gradient of the objective function must be proportional to the gradient of the constraint at the optimal point:

$$\nabla f = \lambda \nabla g$$

This gives us the Lagrangian function $\mathcal{L}(x, y, \lambda) = f(x, y) - \lambda g(x, y)$. We equate the partial derivatives to zero:

1. $\frac{\partial f}{\partial x} = \lambda \frac{\partial g}{\partial x}$
2. $\frac{\partial f}{\partial y} = \lambda \frac{\partial g}{\partial y}$
3. $g(x, y) = 0$ (The constraint itself)

3. Compute Partial Derivatives

- **For x :**

$$\frac{\partial f}{\partial x} = 2x - 2$$

$$\frac{\partial g}{\partial x} = 1$$

$$\text{Equation (1): } 2x - 2 = \lambda(1) \implies \lambda = 2x - 2$$

- **For y :**

$$\frac{\partial f}{\partial y} = 8y + 8$$

$$\frac{\partial g}{\partial y} = 2$$

$$\text{Equation (II): } 8y + 8 = \lambda(2) \implies 4y + 4 = \lambda$$

- **For Constraint:** Equation (III): $x + 2y = 7$

4. Solve the System of Equations

Equate the expressions for λ from (I) and (II):

$$2x - 2 = 4y + 4$$

Rearrange to solve for x in terms of y :

$$2x - 4y = 6$$

Divide by 2:

$$x - 2y = 3 \quad \dots \text{(IV)}$$

Now we have a system of two linear equations with two variables (III and IV):

$$1. \quad x + 2y = 7$$

$$2. \quad x - 2y = 3$$

Add the two equations together to eliminate y :

$$(x + 2y) + (x - 2y) = 7 + 3$$

$$2x = 10$$

$$x = 5$$

Substitute $x = 5$ back into equation (III):

$$5 + 2y = 7$$

$$2y = 2$$

$$y = 1$$

So, the stationary point is $(5, 1)$.

5. Calculate the Minimum Value Now, substitute $x = 5$ and $y = 1$ back into the original objective function $f(x, y)$ to find the minimum value.

$$f(5, 1) = (5)^2 + 4(1)^2 - 2(5) + 8(1)$$

$$= 25 + 4 - 10 + 8$$

$$= 29 - 10 + 8$$

$$= 19 + 8$$

$$= 27$$

Final Answer: The minimum value of the function subject to the constraint is **27**.